

DATA 583

Advanced Predictive Modelling



Likelihood Theory

- **Maximum Likelihood Estimation**
- **Some Likelihood Theory:**
 - **Invariance**
 - **Score Function**
 - **Fisher Information**
 - **Likelihood Ratio Tests**
- **Exponential Family of Distributions**

Regression when the distribution is not normal

- We have used least-squares to estimate parameters of regression models when the responses are normally distributed?
- What if the responses follow a different distribution?
- Least-squares is robust to non-normality, but the likelihood offers a better alternative when we know how to model the responses.
- In this lecture, we will discuss the important properties of the likelihood and log likelihood functions, from a general point of view.

Maximum likelihood estimation - summary

The likelihood function for n independent measurements $y_1, \dots, y_n$ coming from a population modelled by a density $f(y)$ is

$$L(\theta) = f(y_1)f(y_2) \cdots f(y_n)$$

θ denotes parameter(s) to be estimated.

The maximum likelihood estimator for θ is the value which maximizes the likelihood function.

Often, the maximum likelihood estimator can be found by setting the derivative of the log likelihood to 0 and solving for θ .

Likelihood properties — invariance

- Suppose θ is an unknown parameter in a probability model $f(y; \theta)$. Recall that the mle $\hat{\theta}$ is the maximizer of $L(\theta) = f(y; \theta)$.
- Suppose $\gamma = g(\theta)$ for some known 1-1 function $g(\cdot)$. The quantity γ is also a parameter and we can reparametrize the model $f(y)$ in terms of γ instead of θ : $f(y; g^{-1}(\gamma))$.
- Re-define the likelihood function to be

$$L(\gamma) = f(y; g^{-1}(\gamma))$$

Also, define $\hat{\gamma} = g(\hat{\theta})$. Then $\hat{\gamma}$ is the maximum likelihood estimator for γ since, for any γ ,

$$L(\gamma) = f(y; \theta) \leq f(y; \hat{\theta}) = L(\hat{\gamma}).$$

That is $\hat{\gamma}$ is a maximizer of the likelihood when the model is parametrized in terms of γ .

Likelihood properties — invariance

- Invariance prevents ambiguities from occurring which might otherwise arise when changing scales or when taking logs or square roots of parameters.

Invariance – example: normal model

- Suppose y is normally distributed with mean 0 and standard deviation σ . Under this parametrization, the probability density function is

$$f(y) = \frac{e^{-\frac{1}{2}y^2/\sigma^2}}{\sqrt{2\pi}\sigma}.$$

- The likelihood function $L(\sigma) = f(y)$ is maximized at

$$\hat{\sigma} = |y|.$$

- If we estimate the variance $\gamma = \sigma^2$, we obtain

$$\hat{\gamma} = y^2.$$

Example: Poisson distribution: log rate versus rate

- Suppose $y_1, y_2, \dots, y_n$ are a collection of independent counts from a Poisson, λ distribution.
- The MLE for λ is $\bar{y}$, where $\bar{y}$ is the average of the counts.
- By the invariance principle the MLE for $\log(\lambda)$ is $\log(\bar{y})$.

The score function: the derivative of the log likelihood

The expected value and variance of the derivatives of the log likelihood are often useful, particularly, in establishing large sample properties of maximum likelihood estimators.

We will consider a single observation from a probability density function with unknown parameter θ : $f(y; \theta)$.

The log likelihood is given by

$$\ell(\theta) = \log f(y; \theta).$$

If $\ell(\theta)$ is a differentiable function of θ , we can show that under fairly general conditions, the expected value of the first derivative, often called the score function, is 0:

$$E \left[\frac{\partial \ell(\theta)}{\partial \theta} \right] = 0. \tag{1}$$

The expected score function

Equation (1) can be demonstrated by noting that the left hand side of this identity is defined as

$$\int \frac{\partial \ell(\theta)}{\partial \theta} f(y; \theta) dy$$

which, by the definition of the log likelihood, can be written as

$$\int \frac{1}{f(y; \theta)} \frac{\partial f(y; \theta)}{\partial \theta} f(y; \theta) dy = \int \frac{\partial f}{\partial \theta} dy = \frac{\partial}{\partial \theta} \int f(y; \theta) dy = 0.$$

The interchange of differentiation and integration depends on conditions on f as a function of y and θ , which are often, but not always, satisfied in practice.

The importance of the result at (1) cannot be understated: we expect the likelihood to be have a critical point at the true value of θ .

We will see shortly that the critical point is often a maximizer.

Information, variance, and derivatives

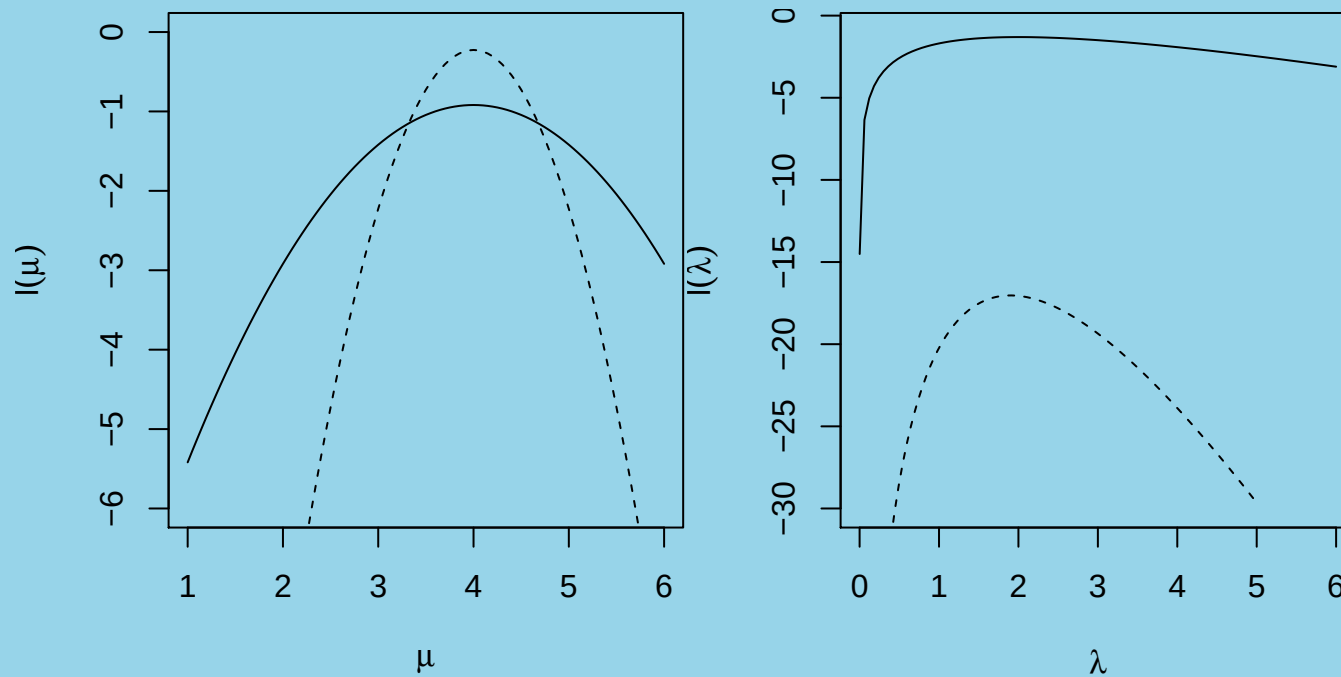
The variance of the score function contains information about the potential for precise estimation of the parameter θ .

When the second derivative of the log likelihood exists, it has a useful relation to this variance, as it contains information about the curvature of the log likelihood in the vicinity of the true value of θ .

If the likelihood is flat, there will be less precision than if the likelihood is more curved.

In other words, a larger second derivative, in magnitude, would give more precision.

Curvature of the likelihood function and precision of estimation



Curvature of the log likelihood function and precision of estimation. Left panel: log likelihood of the normal pdf at $y = 4$, for $\sigma^2 = 1$ (solid curve) and $\sigma^2 = 0.25$ (dashed curve). Right panel: log likelihood for the Poisson distribution with rate parameter λ with one observation at $y = 2$ (solid curve) and with 10 independent observations having coming from a Poisson population with rate $\lambda = 2$.

Information, variance, and derivatives - normal case

- The second derivative of the log likelihood function in the normal case is $-1/\sigma^2$, so smaller values of σ^2 correspond to increased curvature around the maximum.
- When σ is 0.5, there is more curvature than when $\sigma = 1$, and at the same time, there is more certainty about the estimate when σ is smaller.

Information, variance, and derivatives - Poisson case

- In the Poisson case, the second derivative of the log likelihood is $-y/\lambda^2$ when there is a single observation y , and the second derivative is $-n\bar{y}/\lambda^2$ when there are n independent observations.
- The curvature is higher when $n > 1$, and this corresponds to greater precision in estimation about the true value of λ .

Information, variance, and derivatives

- The variance of the first derivative of the log likelihood, evaluated at the true value of the parameter, is referred to as the Fisher Information, $\mathcal{I}$.

•

$$\mathcal{I} = \mathbf{Var} \left(\frac{\partial \ell(\theta)}{\partial \theta} \right) = E \left[\left(\frac{\partial \ell(\theta)}{\partial \theta} \right)^2 \right] \quad (2)$$

- Also, note that

$$E \left(\frac{\partial^2 \ell(\theta)}{\partial \theta^2} \right) = -E \left[\left(\frac{\partial \ell(\theta)}{\partial \theta} \right)^2 \right] \quad (3)$$

Using ratios of likelihoods in statistical testing

Since the likelihood function is expected to take on larger values at parameter values near the true value, and smaller values elsewhere, it is possible to test whether a particular parameter value θ_0 is reasonable for a given data set.

To do this, the likelihood is calculated at θ_0 as well as at the MLE.

The likelihood at the MLE will be larger than the likelihood anywhere else, so the ratio of the two likelihood values should be a number that is at least one.

How much larger this likelihood ratio statistic is than one gives a measure of evidence *against* the assumption that the true parameter value is θ_0 .

Using ratios of likelihoods in statistical testing

The likelihood ratio test of the null hypothesis that $\theta = \theta_0$ proceeds as follows:

1. Calculate the likelihood ratio: $R_L = L(\theta_{\text{MLE}})/L(\theta_0)$
2. Calculate the probability that such a likelihood ratio could exceed R_L for any other sample coming from a population for which the null hypothesis is true. This probability is the p-value for the test.

Since evidence against the null hypothesis increases with the value of the likelihood ratio, R_L , the p-value as defined above provides a corresponding measure of evidence.

Small p-values are indicators of strong evidence against the null hypothesis.

Using ratios of likelihoods in statistical testing

Although the likelihood ratio is the basis for the test, it is often more convenient to work the log of the likelihood ratio.

The likelihood ratio then becomes a difference of log likelihoods:

$$\log(L(\theta_{\mathbf{MLE}})/L(\theta_0)) = \log(L(\theta_{\mathbf{MLE}})) - \log(L(\theta_0)).$$

Example – impurity proportions

A possible model for proportion x of impurity is

$$f(x) = (\alpha + 1)x^\alpha, \quad x \in (0, 1), \quad \alpha > -1.$$

Interest centers on α . We know how to estimate it from one or more independent observations by maximizing the likelihood:

```
# mle:
alphahat <- function(x) {
  -1/mean(log(x)) - 1
}

# examples: x = .37; x = .9; x = .37, .9, .8
alphahat(.37)

## [1] 0.005780954
```

Example – impurity proportions

```
alphahat (.9)
```

```
## [1] 8.491222
```

```
alphahat (c (.37, .9, .8))
```

```
## [1] 1.267991
```

How do we test whether α differs from some possible baseline value α_0 .

For example, $\alpha_0 = 0$ corresponds to the uniform distribution on $(0, 1)$, and we might want to quantify the evidence against the null hypothesis that α is 0.

Example – impurity proportions

The likelihood ratio test offers a way to do this. The test statistic is constructed as

$$R_L = \frac{L(\alpha_{\text{mle}})}{L(\alpha_0)}.$$

For our problem, the likelihood ratio could be calculated as follows, assuming a single observation:

```
LR <- function(x, alpha) {  
  # Here, alpha represents the null hypothesis parameter value  
  (-1/log(x)) * x^(-(1/log(x) + 1)) / ((alpha + 1) * x^alpha)  
}
```

Example – impurity proportions

```
# examples: n = 1 only; null distribution is uniform  
LR(.37, alpha = 0)  
  
## [1] 1.000017  
  
LR(.9, alpha = 0)  
  
## [1] 3.879584
```

We are not yet in a position to say how much evidence we have against the null hypothesis in either of these cases, since we don't know the distribution of the test statistic.

In this case, determining the distribution is somewhat challenging, but a parametric bootstrap can be employed.

Example – impurity proportions

First, we can construct a random number generator for X values coming from the model pdf $f(x)$:

```
rX <- function(n, alpha) (runif(n))^(1/(alpha + 1))

# examples:
rX(5, 0)      # 2 uniform variates

## [1] 0.83590574 0.22461975 0.05679458 0.21354426 0.92340140

rX(3, 2)      # 3 variates with alpha = 2

## [1] 0.8725228 0.5442155 0.9740518
```

Example – impurity proportions

Using such values, we can simulate random values of the likelihood ratio as follows:

```
rLR <- function(n, alpha) {
  x1 <- rX(n, alpha)
  LR(x1, alpha = 0)
}
# examples:
rLR(5, 0) # 5 LR values (corresp. to n = 1); alpha = 0

## [1] 1.136797 43.245267 4.272020 1.006806 1.196522

rLR(3, 2) # 3 LR values (corresp. to n = 1); alpha = 2

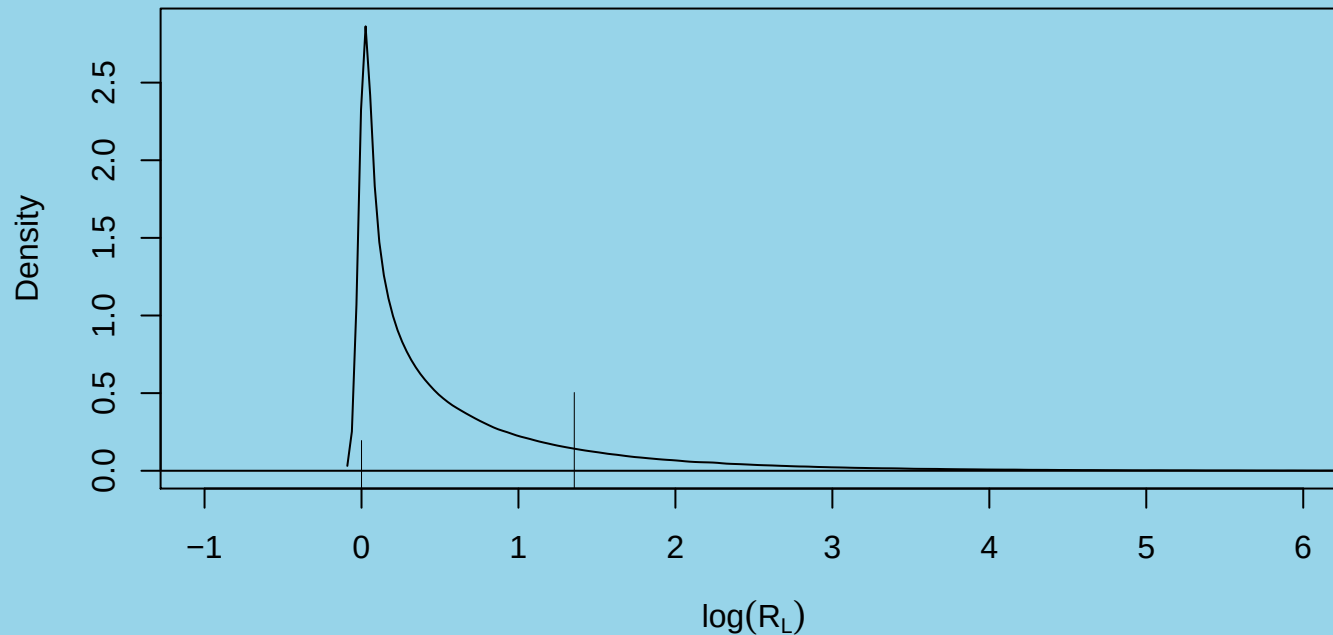
## [1] 13.661378 5.748507 4.913746
```


Example – impurity proportions

Simulating the distribution under the null hypothesis (uniform distribution) proceeds as follows:

```
y <- rLR(1000000, alpha = 0)
plot(density(log(y)), xlim=c(-1, 6), main = "",
     xlab = expression(log(R[L]))) # this estimates the density
# from the logs of the simulated data
rug(log(LR(.37, 0)), ticksize = .1) # this shows the evidence
# against the null hypothesis if x = .37
rug(log(LR(.9, 0)), ticksize = .2)  # this shows the evidence
# against the null hypothesis if x = .9
abline(h = 0) # a horizontal base line
```

Example – impurity proportions



This shows an estimate of the probability density function of the log of the likelihood ratios, for testing the hypothesis that $\alpha = 0$ based on a single observation.

The p-values for the two tests correspond to the areas under the density curve to the right of the ticks for the rug plot.

It is now clear that there is no evidence against the null hypothesis when $x = .37$ (shorter tick), but when $x = .9$, there is fairly strong evidence against the null hypothesis (that $\alpha = 0$).

Multiple observations and an asymptotic χ^2 distribution

- For independent and normally distributed data, $2 \log(R_L)$ has a χ^2 distribution on 1 degree of freedom for any sample size.
- If we have n independent observations from a non-normal population, it can be shown that $2 \log(R_L)$ has a limiting distribution which is χ^2 on 1 degree of freedom as the sample size n increases, under the assumption that the null hypothesis is true.
- This result can be used to determine approximate p-values.

Example – homogeneous Poisson model

- In a sample of 7 rats who received 1 injection of carcinogen which was allowed to incubate for 6 weeks, the numbers of single multiplicity aberrant crypt foci (ACF) in the proximal section of the colon were found to be

```
counts <- c(11, 1, 2, 4, 4, 6, 0)
```

- Assuming a Poisson distribution, we estimate the parameter λ to be 4.

Likelihood ratio testing in the homogeneous Poisson model

- If we want to test the hypothesis that the true mean number of single crypt ACF is really 1, we can employ the likelihood ratio test with $\lambda_0 = 1$:

$$2 \log(R_L) = 2(-n\lambda_0 + \sum_{i=1}^n x_i \log(\lambda_0) + n\lambda_{\text{mle}} -$$

$$\sum_{i=1}^n x_i \log(\lambda_{\text{mle}}))$$

$$= -2n(\lambda_0 - \lambda_{\text{mle}}) - 2 \log(\lambda/\lambda_0) \sum_{i=1}^n x_i$$

Likelihood ratio testing in the homogeneous Poisson model

- **Plugging in the data, we have**

$$= 14(1 - 4) + 2(\log(4/1) \times 28 = 35.6$$

Comparing with the χ^2 distribution on 1 degree freedom, we have a p-value of 2.3828076×10^{-9} .

- **This means we have very strong evidence that the true parameter is not 1.**

The exponential family of distributions

- A probability density function $f(x)$ is said to belong to the exponential family of distributions when

$$f(x) = \mathbf{e}^{(x\theta - b(\theta))/a(\phi) + c(x, \phi)}$$

for some functions a , b and c . The parameter θ is referred to as the canonical parameter.

- The parameter ϕ is often referred to as the dispersion parameter.
- Exponential family distributions are very common.

Example – normal

- The normal distribution is an example:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \mathbf{e}^{-\frac{1}{2\sigma^2}(x-\mu)^2}.$$

- The canonical parameter is $\theta = \mu$, $b(\theta) = \mu^2/2$, $\phi = \sigma^2$, $a(\phi) = \sigma^2$, and $c(x, \phi) = -x^2/(2\phi) - \log(\sqrt{2\pi\phi})$.

Example – Poisson

- Another example is the Poisson distribution

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}.$$

- This time, the canonical parameter is $\theta = \log(\lambda)$, $b(\theta) = \lambda = e^\theta$, $\phi = 1$, $a(\phi) = 1$, and $c(x, \phi) = -\log(x!)$.

Other examples include the Bernoulli distribution, binomial distribution, gamma distribution, and geometric distribution.

Notable examples of distributions that are not members of the exponential family are the uniform distribution and other beta distributions.

Expectation

- Differentiating the log likelihood

$$\ell(\theta) = \log(f(x)) = (x\theta - b(\theta))/a(\phi) + c(x, \phi)$$

with respect to θ , we have

$$\frac{\partial \ell(\theta)}{\partial \theta} = (x - b'(\theta))/a(\phi).$$

Taking expectations on both sides, and remembering that the expected value of the score function is 0, we can obtain

$$b'(\theta) = E[x].$$

- The inverse of the function $b'(\theta)$ is referred to as the canonical link function since it links the canonical parameter θ with the expectation of the random variable in question.

Variance

Taking a second derivative of the log likelihood function with respect to θ yields

$$\frac{\partial^2 \ell}{\partial \theta^2} = \frac{-b''(\theta)}{a(\phi)}.$$

Taking expectations on both sides of this equation, and recalling

$$E \left[\frac{\partial^2 \ell}{\partial \theta^2} \right] = -E \left[\left(\frac{\partial \ell}{\partial \theta} \right)^2 \right] = -E[(x - b'(\theta))^2 / a^2(\phi)] = -\mathbf{Var}(x) / a^2(\phi),$$

we see that

$$\mathbf{Var}(x) = b''(\theta) a(\phi).$$

Variance

Briefly consider the Poisson distribution as an example here.

We saw that $b(\theta) = e^\theta = \lambda$.

Differentiating twice, with respect to θ , we have $b''(\theta) = e^\theta = \lambda$.

Since $a(\phi) = 1$, we have $\text{Var}(x) = \lambda$, as is well known.

Maximum likelihood estimation within the exponential family

We begin by fitting an exponential family model $f(y; \theta)$ when there is a single response value y .

The log likelihood is given by

$$\ell(\theta) = \log(f(y; \theta)) = \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi).$$

The maximum likelihood estimator for θ can be obtained by differentiating with respect to θ

$$\ell'(\theta) = \frac{y - b'(\theta)}{a(\phi)}$$

and finding the solution of $\ell'(\theta) = 0$.

In other words, we must solve the equation

$$b'(\theta) = y$$

for θ .

Example – homogeneous Poisson model

If y follows a Poisson distribution with mean λ , the log likelihood is given by

$$\ell(\lambda) = y \log(\lambda) - \lambda - \log(y!)$$

so we see that $\theta = \log(\lambda)$, $a(\phi) = 1$, $c(y, \phi) = -\log(y!)$ and $b(\theta) = \lambda = e^\theta$. In terms of θ , the log likelihood is expressed as

$$\ell(\theta) = y\theta - e^\theta - \log(y!).$$

Differentiating $\ell(\theta)$ with respect to θ , we see that we must solve

$$b'(\theta) = e^\theta = y$$

for θ :

$$\hat{\theta} = \log(y)$$

and by invariance of maximum likelihood estimation, we also have

$$\hat{\lambda} = e^{\hat{\theta}} = y.$$

Example – homogeneous Bernoulli model

If y follows a Bernoulli distribution with mean p , the log likelihood is given by

$$\ell(p) = y \log(p) + (1 - y) \log(1 - p) = y \mathbf{logit}(p) + \log(1 - p)$$

so $\theta = \mathbf{logit}(p)$, $a(\phi) = 1$, $c(y, \phi) = 0$, and $b(\theta) = \log(1 + e^\theta)$.

$$\rightsquigarrow \ell(\theta) = y\theta - b(\theta); \quad p = e^\theta / (1 + e^\theta).$$

Differentiating ℓ as a function of θ with respect to θ , our estimating equation for θ is

$$b'(\theta) = \frac{e^\theta}{1 + e^\theta} = y$$

which does not have a very useful solution, since y only takes on values 0 or 1. More data points are needed to make any progress with this model.

Estimation from samples of observations

When we have multiple response values, $y_1, y_2, \dots, y_n$ where each $y_j \sim f(y_j; \theta_j)$, we can set up the log likelihood for each data point as done earlier in this section, and we find that the maximum likelihood estimator for θ_j is

$$\hat{\theta}_j = b'^{-1}(y_j)$$

for $j = 1, 2, \dots, n$.

This type of model for the response values is referred to as a saturated model.

It is impossible to construct a more flexible model for the given data than this, using likelihood considerations.

The saturated model is very flexible and allows the closest fit to the data.

When this model does not fit the data well, the implication is that the assumed underlying probability model is incorrect.

Estimation from samples of observations

Observe that we have not had to assume independence in order to do the fitting; this is analogous, and indeed generalizes what we observed earlier about fitting models via least-squares.

Independence is not needed to do the parameter estimation, but statistical inference is not possible without inducing some restrictions on the form of dependence in the data.

Practical modeling also requires restrictions on the form of the model.

We consider the most restrictive form next.

Models for independent and identically distributed observations

Given independent and identically distributed responses $y_1, y_2, \dots, y_n$ (i.e. a random sample) from a probability density function in the exponential family, governed by parameter θ , the log likelihood is given by

$$\ell(\theta) = \sum_{i=1}^n \log(f(y_i; \theta)) = \sum_{i=1}^n \frac{y_i \theta - b(\theta)}{a(\phi)} + \sum_{i=1}^n c(y_i, \phi).$$

The maximum likelihood estimator for θ can once again be obtained by differentiating with respect to θ

$$\ell'(\theta) = \sum_{i=1}^n \frac{y_i - b'(\theta)}{a(\phi)}$$

and finding the solution of $\ell'(\theta) = 0$. In other words, we must solve the equation

$$b'(\theta) = \frac{1}{n} \sum_{i=1}^n y_i = \bar{y}$$

for θ .

Example – homogeneous Poisson model

For n independent observations from a Poisson distribution with mean λ , the log likelihood is given by

$$\ell(\lambda) = \sum_{i=1}^n y_i \log(\lambda) - n\lambda - \sum_{i=1}^n \log(y_i!).$$

Take $\theta = \log(\lambda)$, and $b(\theta) = e^{\theta}$. In terms of θ , the log likelihood becomes

$$\ell(\theta) = \sum_{i=1}^n y_i \theta - n e^{\theta} - \sum_{i=1}^n \log(y_i!).$$

Differentiating this with respect to θ , we must solve

$$n e^{\theta} = \sum_{i=1}^n y_i \text{ for } \theta.$$

$$\hat{\theta} = \log\left(\sum_{i=1}^n y_i / n\right) = \log(\bar{y})$$

and by invariance of maximum likelihood estimation, we also have $\hat{\lambda} = e^{\hat{\theta}} = \bar{y}$.

Example – homogeneous Bernoulli model

If the responses follow a Bernoulli distribution with mean p , the log likelihood is given by

$$\ell(p) = \sum_{i=1}^n y_i \log(p) + \sum_{i=1}^n (1 - y_i) \log(1 - p) = \sum_{i=1}^n y_i \mathbf{logit}(p) + n \log(1 - p)$$

and with $\theta = \mathbf{logit}(p)$, the log likelihood can be expressed in terms of theta as

$$\ell(\theta) = \sum_{i=1}^n y_i \theta + n \log(1 - p(\theta))$$

where $p(\theta) = \mathbf{e}^\theta / (1 + \mathbf{e}^\theta)$. Differentiating the log likelihood with respect to θ , our estimating equation for θ is

$$\frac{\mathbf{e}^\theta}{1 + \mathbf{e}^\theta} = \frac{1}{n} \sum_{i=1}^n y_i = \bar{y}$$

which is the proportion of 1's in the observed sample.
